

NAME: NEPTUN-CODE: MAT or PHYS?:

Probability theory exam, 2025. dec. 18.

*Working time: 100 minutes. Non-programmable, offline calculator can be used,
standard normal distribution table on the other side.*

Maximum possible score (including bonus): 110 points, but even 100 points is considered as 100%.

- The. 1.** (a) (4 points) Define the expected value and standard deviation of discrete probability distributions.
(b) (4 points) Give an example of a distribution with a finite expected value but an infinite variance.
(c) (4 points) There are two formulas for the standard deviation (one: the squared distance from the expected value; the other: expressed in terms of the first and second moments). Write down the formula that you did not write down in task (a) and prove that the two formulas give the same value.
(d) (4 points) Derive the formula for the standard deviation of the binomial distribution.
- The. 2.** Let X and Y be independent absolutely continuous random variables. Let $Z = X + Y$. Derive the formula for the distribution function of Z (6 points). From this, derive the formula for the density function of Z (3 points). If X and Y are both uniformly distributed on the interval $[2, 4]$, then give the formula for the density function of Z and draw the graph of the density function of Z (4 points), and calculate the probability $\mathbb{P}(Z > 7)$ (4 points).
- The. 3.** Let (X, Y) be jointly absolutely continuous. Define the conditional expectation of X under the condition $Y = y$ (3 points), and the conditional expectation of $\mathbb{E}(X | Y)$ (2 points). State and prove the tower rule of conditional expectation (5 points). Define the meaning of the terms in the conditional variance formula (3 points) and prove the formula (4 points).
- Pr1. 1.** Gaspard carries 3 pices of gold, 2 frankincense and 1 myrrh in his gift package, Melchior carries 2 pieces of gold, 1 frankincense and 3 myrrh in his package, and Balthazar carries 1 piece of gold, 3 frankincense and 2 myrrh in his package. The three Wise Men arrived at the Holy Family in random order. Mary blindly reaches into the gift package of the wise man who arrived first and takes out two gifts.
(a) (6 points) What is the expected number of myrrhs in Mary's hand? *Hint:* It's easier if you use the method of decomposition as a sum of indicators!
(b) (6 points) Assuming that Mary holds a piece of gold and a piece of frankincense, what is the probability that Gaspar arrived first?
(c) (5 points) Assuming that she holds a piece gold and a piece of frankincense, what is the expected number of myrrhs in the same package?
- Pr2. 2.** (16 points) We roll two independent fair dice. What is the joint probability weight function of X and Y if X is the minimum of the two throws and Y is their sum?
- Pr3. 3.** On average, the Tume parcel delivery company receives an online Christmas gift order every two seconds. Independently of this, the rival Azla parcel delivery company also receives an order every two seconds on average.
(a) (4 points) What is the variance of the time of the fifth order received by Azla after 10.00am?
(b) (5 points) What is the variance of the number of orders received by both companies together in the next 10 seconds?
(c) (8 points) Estimate the probability that Tume will receive at least 50 more orders than Azla in the next hour. *Instructions:* Φ table on the opposite page.
- Bonus** (10 points) Estimate the probability of the joint occurrence of events that Azla will receive more orders between noon and 6 pm today and also between noon and midnight today. *Instructions:* Φ table on the opposite page.

	$\Phi(z)$									
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000